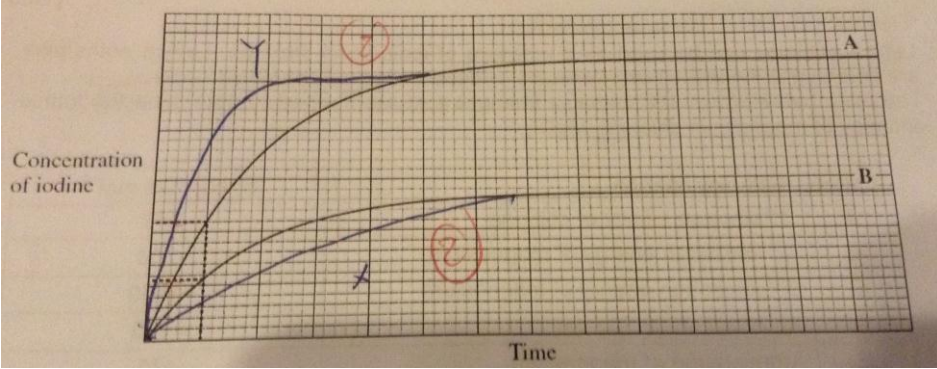
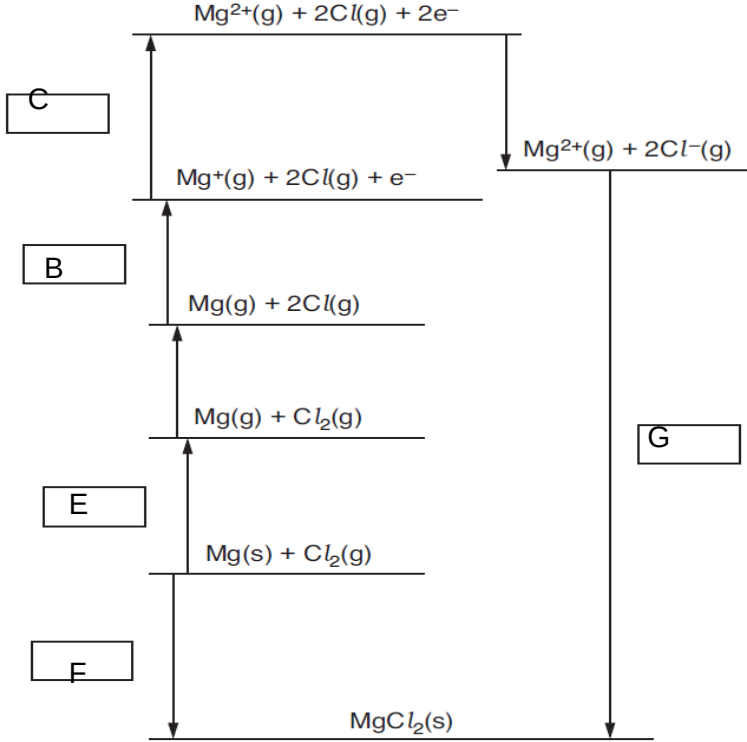


2013 NJC H1 Prelims Paper 2 Answers

1(a)	$\text{Cr}^{3+}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$ $\text{Mn}^{2+}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$		
(b)	(i)	$\text{CH}_3\text{CH}_2\text{OH}$ $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	
	(ii)	I: Orange to green II: No. of moles of $\text{K}_2\text{Cr}_2\text{O}_7 = \frac{47.2}{39.1 \times 2 + 52 \times 2 + 16 \times 7}$ = 0.160 III: No. of moles of $\text{C}_3\text{H}_6\text{O} = \frac{18}{60}$ = 0.3 $\text{C}_3\text{H}_6\text{O}$ is the limiting reagent. Theoretical mass of $\text{C}_3\text{H}_6\text{O}$ produced = $0.3 \times (36 + 6 + 16)$ = 17.4g % yield = $\frac{5.22}{17.4} \times 100\%$ = 30%	
	(iii)	I: Impurity – carboxylic acid Reason – Chemical shift of 9.0-13.0ppm represents the presence Of R-COOH group. II: $\text{CH}_3\text{CH}_2\text{OH}$	

2(a)	Order: 1 st order Explanation: The same amount of time taken gives twice the concentration for curve A when compared to curve B.	
(b) (c)		
(d)	Reaction between I^- and $\text{S}_2\text{O}_8^{2-}$ has high activation energy as they are of the same charge and energy is required to overcome the repulsive forces.	

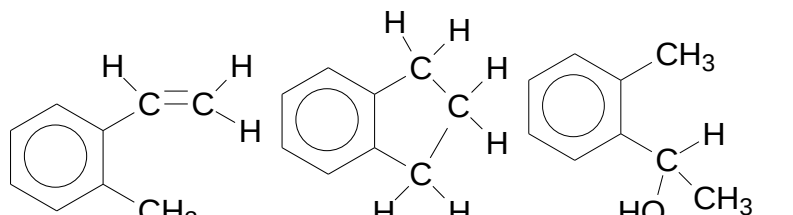
3(a)		
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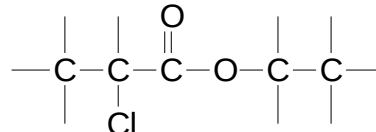
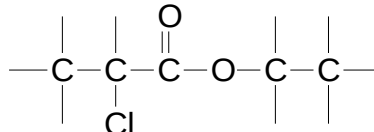
(b)	$(1450+736+2 \times 150+76+642) - (2 \times 349)$ $= -2506 \text{ kJmol}^{-1}$	
(c)	$ L.E \propto \frac{q^+ \times q^-}{r^+ + r^-}$ Since $r_{Cl^-} > r_{F^-}$, therefore the lattice energy of magnesium fluoride is more exothermic than magnesium chloride. Hence more energy is required to break the ionic bond between magnesium ion and fluoride ion.	

4(a)	C_4H_8O		
(b)	(i)	Test 3: alkene	
	(ii)	Test 4: alcohol	
(c)	(i)	Carbonyl	
	(ii)	1° alcohol	
(d)		Free rotation about the bond is restricted by the presence of multiple bonds. The carbon atoms on both ends of the C=C double bond contain two non-identical groups.	
(e)		$\begin{array}{c} H_3C \quad \quad CH_2OH \\ \diagdown \quad \diagup \\ C = C \\ \diagup \quad \diagdown \\ H \quad \quad H \end{array}$ Cis $\begin{array}{c} H \quad \quad CH_2OH \\ \diagdown \quad \diagup \\ C = C \\ \diagup \quad \diagdown \\ H_3C \quad \quad H \end{array}$ trans	

5(a)	(i)	-2	
	(ii)	A: HCHO B: HCOOH C: CO ₂	
	(iii)	A: K ₂ Cr ₂ O ₇ , dilute sulfuric acid and heat with distillation B: K ₂ Cr ₂ O ₇ or KMnO ₄ dilute sulfuric acid and heat C: Heat with excess hydrogen or KMnO ₄ dilute sulfuric acid and heat	
(b)	(i)	$Q_{50\%} = 100 \times 4.18 \times 30$ $= 12\,540\text{J}$ $n_{\text{ethanol}} = 1.5/46 = 0.0326 \text{ moles}$ $\Delta H = -Q_{100\%}/n_{\text{ethanol}}$ $= -769 \text{ kJmol}^{-1}$	
	(ii)	There will be heat lost to the surrounding.	
(c)	(i)	$K_c = [\text{CH}_3\text{OH}]/[\text{H}_2]^2[\text{CO}] \text{ mol}^{-2}\text{dm}^6$	
	(ii)	$K_c = (0.8/1.5) / (0.4/1.5)^2(0.4/1.5)$ $= 56.25$	
	(iii)	$[2(436)+1077]-[3(410)+360+460]$ $\Delta H = -101 \text{ kJmol}^{-1}$	
(d)	(i)	D: CH ₃ CH(CH ₃)CH ₂ OH Primary	
	(ii)	E: CH ₃ CH ₂ CH ₂ CH ₂ OH Primary F: CH ₃ CH ₂ CH(OH)CH ₃ Secondary G: (CH ₃) ₃ COH Tertiary	
	(iii)	F	
	(iv)	F	

6(a)	The total number of protons and neutron in the nucleus		
(b)	Deflection to the positive plate The angle of deflection is greater.		
(c)	(i)		
	(ii)	Bond angle of O₃ is between 110° to 120° because lone pair-bond pair repulsion is greater than bond pair-bond pair repulsion. The bond angle of NO₂ is between the bond angle of O₃ to 170° as a single unpaired electron will be less effective in diminishing the basic angle, and so the bond angle should be greater than that for O₃.	

	(iii)	Chlorine is able to expand its octet of electrons through the 3d orbital which is accessible for electron occupation. Fluorine cannot expand its octet and is unable to form dative bond as it is too electronegative.	
(d)	(i)	For magnesium to react with oxygen it is necessary to apply heat, ideally by holding a piece of the metal in tongs in an atmosphere of oxygen, such as in a gas jar. Magnesium burns on heating with a brilliant white flame.	
	(ii)	NaCl: dissolves but do not under go hydrolysis MgCl ₂ : undergoes partial hydrolysis AlCl ₃ : Undergoes hydrolysis SiCl ₄ : Undergoes hydrolysis and release HCl PCl ₅ : Undergoes hydrolysis and release HCl	
(e)	(i)	No. of moles of S ₂ O ₃ ²⁻ = 0.0015 No. of moles of I ₂ = 0.00075	
	(ii)	No. of moles of O ₃ = 0.00075 Volume of O ₃ = 16.8 cm ³ % of O ₃ present = 3.36%	
(f)	(i)	Aldehyde: K ₂ Cr ₂ O ₇ , dilute H ₂ SO ₄ , heat with immediate distillation Carboxylic acid: K ₂ Cr ₂ O ₇ /KMnO ₄ dilute H ₂ SO ₄ , heat	
	(ii)	 L M N	

7(a)	(i)	NaCN, dilute H ₂ SO ₄ , trace amount of NaCN	
	(ii)	Dilute H ₂ SO ₄ , heat	
(b)	(i)	Assuming lactic acid is a strong acid, pH = -lg[H ⁺] = 0.699 Since the pH of lactic acid is 1.7, it will be a weak acid as it is only partially dissociated.	
(c)	(i)	It is a solution that will resist pH changes when small amount of acid or alkali is added. $\text{CH}_3\text{CH}(\text{OH})\text{COOH} + \text{OH}^- \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{COO}^- + \text{H}_2\text{O}$ $\text{CH}_3\text{CH}(\text{OH})\text{COO}^- + \text{H}^+ \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{COOH}$	
	(ii)	Phenolphthalein, colourless to pink	
(d)	(i)	Ester	
	(ii)	Hydrolysis	
(e)	(i)	NaOH(aq), heat	
	(ii)	 K  L	

		The COCl is more reactive than C-Cl due to extra electronegative oxygen in COCl will polarise the C-Cl bond to a greater extent which make it easier to break and hence more reactive.	
(f)	1) Test: $\text{Na}_2\text{CO}_3(\text{aq})$ Observation: N – effervescence M – no effervescence 2) Test: 2, 4-dinitrophenylhydrazine Observation: N – no orange precipitate M – orange precipitate 3) Test: $[\text{Ag}(\text{NH}_3)_2]^+$ Observation: N – no silver mirror M – silver mirror 4) Test: Alkaline solution of copper(II) tartarate Observation: N – Blue solution M – reddish brown precipitate		